

MA6.301 (Monsoon 2026)

Probability and Statistics

Assignment 2

Distributions, Expectation, and Joint Random Variables

Instructor: Naresh Manwani

Deadline: TBA

Total Marks: 100 (10% of the course)

Instructions

1. Attempt all questions. Marks for each part are indicated in brackets.
2. Show your reasoning, including relevant probability models, summations, and integration limits. Unsupported numerical answers will not receive full credit.
3. Give exact answers unless a numerical approximation is requested. Report numerical probabilities to **four decimal places**, except where otherwise specified.
4. A calculator, standard normal table, or statistical software may be used for numerical evaluation. Simulation is not a substitute for an analytical solution.
5. Results established in lectures may be quoted unless a question explicitly asks you to derive or prove them.
6. State the support of every PMF or PDF you obtain. Give CDFs for **all real arguments**, including values outside the support.
7. Throughout this assignment:
 - Geometric(p) counts trials **up to and including the first success**.
 - $\text{Exp}(\lambda)$ uses the rate parameter λ .
 - In $\mathcal{N}(\mu, \sigma^2)$, the second parameter is the **variance**.
 - Write $G_X(t) = \mathbb{E}[t^X]$ for a probability generating function, and $\Phi(z)$ for the standard normal CDF.

Suggested references

- Hossein Pishro-Nik, *Introduction to Probability, Statistics, and Random Processes*, available at ProbabilityCourse.com: sections on random variables, expectation, conditional distributions, and joint distributions.
- Dimitri P. Bertsekas and John N. Tsitsiklis, *Introduction to Probability*, 2nd ed., Athena Scientific, 2008: chapters on discrete and general random variables.

The questions below are original exercises aligned with the supplied syllabus, rather than reproductions of exercises from these references.

Question 1. [12 marks]

Geometric waiting time with a retry limit

A device makes successive transmission attempts. Each attempt succeeds with probability $p \in (0, 1)$, independently of all other attempts.

Let X be the number of attempts required for the first success under an unlimited-attempt policy. A practical policy allows at most m attempts, where $m \geq 2$ is an integer. Thus, the actual number of attempts made under this policy is

$$T = \min(X, m).$$

The transmission succeeds under the practical policy exactly when $X \leq m$.

(a) [2] Starting from the geometric PMF or tail probability, prove that

$$\mathbb{P}(X > r + s \mid X > r) = \mathbb{P}(X > s)$$

for all nonnegative integers r, s . Give a one-sentence interpretation in the transmission setting.

(b) [4] Using the partition $X = 1, X > 1$, total expectation, and memorylessness, derive

$$\mathbb{E}[X], \quad \mathbb{E}[X^2], \quad \text{Var}(X).$$

You may assume these moments are finite. Do not quote their known formulas or evaluate the corresponding infinite moment sums directly.

(c) [3] Find the PMF and CDF of T . Pay particular attention to the probability at $T = m$. You may use $\lfloor t \rfloor$, the greatest integer not exceeding t , when writing the CDF.

(d) [3] Use memorylessness and total expectation to obtain a closed-form expression for $\mathbb{E}[T]$. Also find

$$\mathbb{P}(X \leq m \mid T = m).$$

Explain briefly why making exactly m attempts does not necessarily mean that the transmission failed.

Question 2. [10 marks]

Combining batches and approximating rare defects

A manufacturer produces two batches containing 800 and 1200 components, respectively. Every component is defective with probability 0.001, independently of all other components.

Let D_1 and D_2 be the numbers of defective components in the two batches, and let

$$D = D_1 + D_2.$$

- (a) [3] Starting from the binomial PMF, derive $G_{D_1}(t)$ and $G_{D_2}(t)$. Use independence and the product rule for PGFs to obtain $G_D(t)$, and identify the exact distribution of D .
- (b) [3] Find an exact expression for

$$\mathbb{P}(D \geq 2 \mid D \geq 1).$$

Your final expression should not contain an unevaluated sum.

- (c) [4] Approximate the probability in part (b) using a Poisson distribution.

- State the Poisson parameter and explain why this approximation is appropriate.
- Evaluate both the exact probability and the Poisson approximation to **six decimal places**.
- Report the absolute approximation error

$$|\text{Poisson approximation} - \text{exact probability}|$$

to six decimal places.

Question 3. [12 marks]

Independent Poisson counts conditioned on their total

During a fixed observation period, three independent sources generate event counts

$$X_1 \sim \text{Poisson}(2), \quad X_2 \sim \text{Poisson}(3), \quad X_3 \sim \text{Poisson}(1).$$

Let

$$N = X_1 + X_2 + X_3.$$

(a) [3] Derive the PGF of a $\text{Poisson}(\lambda)$ random variable directly from its PMF. Then use PGFs to identify the distribution of N .

(b) [4] For an integer $n \geq 1$, derive

$$\mathbb{P}(X_1 = x_1, X_2 = x_2, X_3 = x_3 \mid N = n).$$

State all conditions on x_1, x_2, x_3 , including when this conditional probability is zero. Identify the resulting multinomial distribution and its parameters.

(c) [2] Use part (b) to identify the conditional distribution of X_1 given $N = 6$. Hence find

$$\mathbb{P}(X_1 = 2 \mid N = 6) \quad \text{and} \quad \mathbb{E}[X_1 \mid N = 6].$$

(d) [3] Are X_1 and X_2 conditionally independent given $N = 6$? Justify your answer by exhibiting specific values a, b for which the conditional joint probability can be compared with the product of the corresponding conditional marginal probabilities. Merely stating that “the total is fixed” is not sufficient.

Question 4. [8 marks]**A sum of binomials with different success probabilities**

Let

$$X \sim \text{Binomial}\left(2, \frac{1}{2}\right), \quad Y \sim \text{Binomial}\left(2, \frac{1}{4}\right),$$

and suppose X and Y are independent. Define $S = X + Y$.

- (a) [3] Use discrete convolution to obtain the complete PMF of S . Show the convolution formula before evaluating the probabilities.
- (b) [2] Calculate $\mathbb{E}[S]$ and $\text{Var}(S)$, using the means and variances of X and Y .
- (c) [3] Determine whether there exists $r \in [0, 1]$ such that $S \sim \text{Binomial}(4, r)$. Give a proof using the mean and variance from part (b). Your answer should explain why independence alone does not guarantee that a sum of binomial random variables is binomial.
-

Question 5. [10 marks]**Dependence that is not detected by a product of expectations**

The joint PMF of X and Y is

$$p_{X,Y}(x, y) = \begin{cases} c(x^2 + y^2), & x, y \in \{-1, 0, 1\}, \\ 0, & \text{otherwise,} \end{cases}$$

where c is a positive constant.

- (a) [2] Find c , and obtain the marginal PMFs of X and Y .
- (b) [3] Find the conditional PMF of X given $Y = 0$. Use it to determine whether X and Y are independent. Support your conclusion with an explicit probability comparison.
- (c) [3] Calculate

$$\mathbb{E}[X], \quad \mathbb{E}[Y], \quad \mathbb{E}[XY].$$

Use your results and part (b) to determine whether $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ is sufficient to establish independence.

- (d) [2] Calculate $\text{Var}(X + Y)$ using

$$\text{Var}(X + Y) = \mathbb{E}[(X + Y)^2] - (\mathbb{E}[X + Y])^2.$$

Compare your answer with $\text{Var}(X) + \text{Var}(Y)$. Does equality of these quantities establish independence in this example?

Question 6. [10 marks]

Empty servers and indicator variables

There are $m \geq 1$ distinguishable jobs and $n \geq 2$ distinguishable servers. Each job is assigned independently and uniformly to one of the n servers.

For $i = 1, \dots, n$, define

$$I_i = \begin{cases} 1, & \text{server } i \text{ receives no jobs,} \\ 0, & \text{otherwise.} \end{cases}$$

Let

$$K = \sum_{i=1}^n I_i$$

be the number of empty servers. Let L be the number of servers receiving exactly one job.

(a) [3] Use indicator variables to derive closed-form expressions for $\mathbb{E}[K]$ and $\mathbb{E}[L]$. Define any additional indicators you introduce.

(b) [2] For distinct servers $i \neq j$, find

$$\mathbb{P}(I_i = 1, I_j = 1).$$

Determine whether I_i and I_j are independent, and justify your conclusion algebraically.

(c) [4] Derive a closed-form expression for $\text{Var}(K)$. Do not assume the indicators are independent.

Hint: Expand K^2 , use $I_i^2 = I_i$, and observe that

$$\mathbb{E}[I_i I_j] = \mathbb{P}(I_i = 1, I_j = 1).$$

(d) [1] Now let $m = n$. Find

$$\lim_{n \rightarrow \infty} \mathbb{E} \left[\frac{n - K}{n} \right],$$

the limiting expected fraction of servers receiving at least one job.

Question 7. [10 marks]

Functions of a uniform random variable

Let $U \sim \text{Uniform}[-2, 4]$.

(a) [2] Write the CDF $F_U(u)$, and calculate

$$\mathbb{P}(-1 < U \leq 2 \mid U > 0).$$

(b) [3] Define $Y = U^2$. Use integration with respect to the density of U to find

$$\mathbb{E}[Y] \quad \text{and} \quad \text{Var}(Y).$$

You are not required to find the density of Y .

(c) [3] Define

$$W = \begin{cases} -1, & U < 0, \\ 0, & 0 \leq U \leq 1, \\ 1, & U > 1. \end{cases}$$

Find the complete PMF of W , its mean, and its variance.

(d) [2] Find the conditional density $f_{U|U>0}(u)$, and use it to calculate $\mathbb{E}[U \mid U > 0]$.

Question 8. [8 marks]

Normal measurements and a decreasing calibration rule

The temperature X , measured in degrees Celsius, satisfies

$$X \sim \mathcal{N}(20, 4).$$

A sensor converts the numerical temperature reading into a dimensionless index

$$Y = 100 - 2X.$$

(a) [2] State the distribution of Y , including its mean, variance, and standard deviation.

(b) [3] Calculate

$$\mathbb{P}(Y \geq 64 \mid X \leq 22).$$

Express the answer in terms of Φ before giving its numerical value. Clearly show how the condition $Y \geq 64$ translates into a condition on X .

(c) [3] Find the unique $d \geq 0$ satisfying

$$\mathbb{P}(|Y - \mathbb{E}[Y]| \leq d) = 0.95.$$

Give d to three decimal places, and state the corresponding central 95% interval for Y .

Question 9. [8 marks]

Exponential lifetime and a capped operating period

The lifetime X , measured in hours, of a component satisfies

$$X \sim \text{Exp}\left(\frac{1}{5}\right).$$

(a) [3] Derive the survival probability $\mathbb{P}(X > x)$ for $x \geq 0$ from the exponential density. Then prove that

$$\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t)$$

for all $s, t \geq 0$.

(b) [2] Given that the component has survived for more than 3 hours, find its conditional density and calculate

$$\mathbb{E}[X \mid X > 3] \quad \text{and} \quad \mathbb{E}[X - 3 \mid X > 3].$$

State which expectation represents the remaining lifetime.

(c) [3] The component is removed from service at failure or at 8 hours, whichever occurs first. Its operating time is therefore

$$T = \min(X, 8).$$

Find the CDF of T and calculate $\mathbb{P}(T = 8)$. Does there exist an ordinary PDF f_T satisfying

$$\mathbb{P}(a < T \leq b) = \int_a^b f_T(t) dt$$

for every $a < b$? Justify your answer using the probability at $T = 8$.

Question 10. [12 marks]

A joint density on a triangular region

The joint PDF of X and Y is

$$f_{X,Y}(x,y) = \begin{cases} c, & 0 < x < y < 2, \\ 0, & \text{otherwise,} \end{cases}$$

where c is a positive constant.

(a) [2] Sketch the support and determine c by normalization.

(b) [3] Find the marginal PDFs $f_X(x)$ and $f_Y(y)$. Determine whether X and Y are independent.

For your independence argument, compare the probability of one explicitly chosen joint event with the product of its marginal event probabilities.

(c) [3] Calculate

$$\mathbb{P}(Y \geq 2X, X + Y \leq 2).$$

Sketch the relevant region and write the integration limits explicitly before evaluating the integral.

(d) [4] Let

$$A = \{Y \leq 1\}.$$

Find

$$\mathbb{P}(A), \quad f_{X|A}(x), \quad \mathbb{E}[X | A].$$

Verify that your conditional density integrates to one.

Hint: To obtain $f_{X|A}$, integrate the joint density over the values of y compatible with both the original support and A , then normalize by $\mathbb{P}(A)$.